

VisDrone-SOT2020: The Vision Meets Drone Single Object Tracking Challenge Results

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Abstract. The Vision Meets Drone (VisDrone2020) Single Object Tracking is the third annual UAV tracking evaluation activity organized by the VisDrone team, in conjunction with European Confer-

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ence on Computer Vision (ECCV 2020). The VisDrone-SOT2020 Challenge presents and discusses the results of 13 participating algorithms in detail. By using ensemble of different trackers trained on several large-scale datasets, the top performer in VisDrone-SOT2020 achieves better results than the counterparts in VisDrone-SOT2018 and VisDrone-SOT2019. The challenging results, collected videos as well as the valuation toolkit are made available at http://aiskyeye.com/. By holding VisDrone-SOT2020 challenge, we hope to provide the community a dedicated platform for developing and evaluating drone-based tracking approaches.

Keywords: Drone-based single object tracking · Drone · Performance evaluation

1 Introduction

Visual object tracking has been one of the most fundamental topics in computer vision with a long list of applications such as video surveillance, intelligent vehicles, and robotics. In the past decades, tracking has been extensively studied and significant progress including stage-of-the-art algorithms [3,5,8,20,31,46,55] and various tracking benchmarks [2,17,26,36,45,59] has been made. However, existing tracking research mainly focuses on video captured by traditional cameras.

Recently, drones equipped with cameras have drawn increasing interest from the tracking community owing to its flexibility. In comparison to videos captured by traditional cameras, the drone videos are more difficult because of abrupt camera motion, extremely small target, frequent view point change in addition to challenges in normal videos. In order to facilitate research of drone-based tracking and its application, it is highly desired to build a dedicated platform with a large amount of drone video sequences in the community. To this end, several attempts have been made by introducing drone tracking benchmarks [14,34,43]. Nevertheless, these datasets only provide performance evaluation and are limited in scale for offering training videos, which motivates the proposal of a large-scale drone tracking benchmark for both training and evaluation of drone-based tracking algorithms.

For this purpose, the VisDrone team has compiled a dedicated large-scale drone tracking benchmark and organized challenges for single-object tracking, *i.e.*, VisDrone-SOT2018 [57] (in conjunction with ECCV 2020) and VisDrone-SOT2019 [16] (in conjunction with ICCV 2019). The VisDrone-SOT2018 consists of 132 videos and discusses 17 submitted trackers. Compared with VisDrone-SOT2018, VisDrone-SOT2019 introduces 35 new sequences and analyzes 22 participating algorithms. To further increase the diversity of drone videos and asses the performance of trackers in the wild, VisDrone-SOT2020 conducts extensive evaluation and discussion of 13 submitted tracking algorithms using the same dataset in VisDrone-SOT2019. We hope that the VisDrone-SOT2020 challenge offers an opportunity for researchers to discuss and share their ideas on

drone-based tracking. All the challenge results, our benchmark and evaluation toolkit can be found at http://aiskyeye.com/.

2 Related Work

Visual object tracking has been extensively studied in recent decades. In this section, we will discuss the related tracking benchmarks and algorithms. More details refer to [67].

2.1 Visual Tracking Benchmark

Benchmarks have played an important role in advancing the research of visual tracking. OTB-2013 [58] is the first benchmark proposed for tracking. It consists of 51 manually annotated sequences with each labeled into 11 attributes. Later, the authors extend OTB-2013 to OTB-2015 [59] by introducing more videos. TC-128 [36] contributes 128 videos, aiming to analyze the impact of color information for tracking. The VOT [29] introduces a series of challenges from 2013 to 2019 with a small-scale of videos for single object tracking. ALOV [50] comprises 314 video sequences with the goal of tracking performance evaluation in various challenging situations. CDBT [39] provides 80 RGB-D videos by focusing the combination of RGB and depth information for tracking. NfS [21] consists of 100 videos with a high frame rate of 240 fps.

In the deep learning era, researchers in the tracking community have resorted to deep features for developing tracking algorithms and achieved state-of-the-art performance. However, the aforementioned datasets limited by their small scales. As a consequence, researchers have been forced to either utilize pre-trained model from ImageNet [13] or adopt video object detection dataset [47,49] for training, which may lead to sub-optimal performance due to domain gap. In order to alleviate this issue, a few large-scale tracking benchmarks have been proposed. LaSOT [17] introduces $1,400$ sequences with more than 3.5M frames for both training and evaluation of tracking algorithms. Each frame is carefully and manually annotated with axis-aligned bounding box. GOT-10k [26] collects around $10,000$ videos for constructing a large-scale benchmark with a focus on short-term tracking. TrackingNet provides 30K videos with each annotated using tracking algorithms. OxUvA [53] collects 366 sequences for long-term tracking and each video is sparsely labeled every 30 frames.

In recent years, drone-based object tracking has drawn extension attention in the community. To boost research on tracking in drone videos, there are several benchmarks. UAV123 [43] collects 123 drone videos with 113K frames for tracking. UAVDT [14] proposes a benchmark with 80K frames for object detection and tracking in drone videos. DTB70 [34] compiles a dataset of 70 drone sequences with 15K frames. Despite facilitating research on drone-based tracking, these benchmarks only provide performance evaluation for drone-based tracking algorithms, which is limited in offering enough videos for training of deep trackers. Tackling this problem, VisDrone-SOT2020 provides a large set of videos for both training and assessment of different trackers.

2.2 Visual Tracking Algorithm

Numerous tracking approaches have been proposed in the past decades. Among these methods, correlation filter based trackers have drawn extension attention owing to its high efficiency. The seminal work MOSSE [6] runs at an impressive speed of several hundred frames per second. KCF [25] extends this method using improved feature representation, achieving remarkable performance. Many later extensions based [6,25] have been done. To handle the problem of scale variation, the approaches of [10,35] leverage an additional scale filter to perform pyramid scale search. The methods in [11,33] propose to enhance correlation filter trackers with complex regularization techniques. Background contextual information is exploited in [22,44] to improve the robustness. Part-based strategy is used in [38] to handle the problem of occlusion for correlation filter tracking. The method in [15] incorporates structural information in local target parts variations using the global constraint in correlation filter tracking.

Inspired by the success of in various vision tasks [23,24,30], deep feature has been adopted for object tracking and demonstrated state-of-the-art results. The approaches in [12,18,41] replaced hand-crafted features with deep features in correlation filter tracking and achieved significant performance gains. To further explore the power of deep features in tracking, end-to-end deep tracking algorithms have been proposed. MDNet [46] introduces an end-to-end classification for tracking. This approach is extended by SANet [19] by exploring structure of target object. VITAL [51] improve the tracking performance by exploiting richer representation for classification. Despite excellent performance, these trackers are inefficient due to extensive deep feature extraction or network fine-tuning. To solve this problem, deep Siamese network has been proposed for tracking [3,52] and achieved a good balance between accuracy and speed. Many extensions based on Siamese tracking have been proposed. Among them, Siamese tracking with regional proposal network (RPN) [20,32,55,66,68] exhibits impressive results and fast speed owing to the effectiveness of RPN in dealing with scale changes. In addition to these methods, another trend is to combine discriminative localization model with a separate scale estimation model for tracking. ATOM [8] utilizes correlation filter method for target localization and an IoUNet [27] for scale estimation. Based on ATOM [8], DiMP [5] incorporates contextual background information for improvement.

2.3 Evaluation Metric

Performance evaluation is crucial for fair comparison of different trackers. There are many evaluation metrics proposed in recent years. The two most common evaluation metrics, precision and success plots, are introduced in [58] by measuring the distance and overlap of tracking result with groundtruth, respectively. These two metrics have been adopted by many later tracking benchmarks [17,36,45,59]. To address the scale sensitivity of precision plot to image scale, the work of [45] suggests an additional normalization precision plot. Different from these methods, the VOT challenge [29] adopts accuracy and robust-

Table 1. Comparison of VisDrone-SOT2020 with existing drone tracking benchmarks. The "Eva." and "Tra." in last column indicate evaluation and training, respectively.

Benchmark	Videos	Mean frames	Total frames	Attributes	Aim
UAV123 [43]	123	915	113K	12	Eva
UAV20L [43]	20	2,934	59K	12	Eva
DTB70 [34]	70	214	15K	12	Eva
UAVDT [14]	100	800	80K	n/a	Tra./Eva
VisDrone-SOT2018 [57]	132	803	106K	12	Tra./Eva
VisDrone-SOT2019 [16]	167	1,131	189K	12	Tra./Eva.
VisDrone-SOT2020	167	1,131	189K	12	Tra./Eva



Fig. 1. Example sequences and annotations of VisDrone-SOT2020. Best viewed in color and zooming-in. (Color figure online)

ness ranking mechanism for tracking performance evaluation. Average overlap is employeed in [26] for comparing different tracking approaches.

3 The VisDrone-SOT2020 Challenge

As discussed above, a large-scale dedicated platform for drone-based tracking is desired. Addressing this issue, VisDrone-SOT2020 introduce a drone tracking benchmark, aiming to provide both training and evaluation for drone-based tracking algorithms.

3.1 Dataset

In the VisDrone-SOT2020 Challenge, we collect 167 drone videos with more than 189K frames. Following VisDrone-SOT2019 [16], the VisDrone-SOT2020 dataset

is divided into three subsets, including *training* set containing 86 videos with 70K frames, *validation* set containing 11 videos with 7K frames and *testing* set containing 95 videos with 145K frames. Compared to VisDrone-SOT2018 dataset with 132 videos and VisDrone-SOT2019 dataset with 167 videos, VisDrone-SOT2020 dataset introduce more challenging drone sequences. To reliably reflect the performance of tracking algorithms *in the wild*, the video sequences of these three subsets are captured from different scenarios with various drone platforms under varying conditions (*e.g.*, the weather and lighting conditions are different in videos of three subsets). In order to ensure the high quality of VisDrone-SOT2020 dataset, each frame is manually annotated with a bounding box. Moreover, in order to allow further performance analysis, we label each video sequence with 12 attributes, including *aspect ratio change* (ARC), *background clutter* (BC), *camera motion* (CM), *fast motion* (FM), *full occlusion* (FOC), *partial occlusion* (POC), *illumination variation* (IV), *low resolution* (LR), *out-of-view* (OV), *similar object* (SO), *scale variation* (SV) and *viewpoint change* (VC). Table 1 summarizes our VisDrone-SOT2020 dataset and comparisons with other drone tracking benchmarks. Figure 1 shows several example videos and annotations of VisDrone-SOT2020.

3.2 Evaluation Metric

Following [58], we adopt precision (PRE) and success (SUC) scores in one-pass evaluation (OPE) to assess different tracking approaches. The PRE score demonstrates the percentage of frames whose estimated average center location errors are within a given threshold distance (*e.g.*, 20 pixels) to the groundtruth. The SUC score is defined as the percentage of frames whose overlaps with groundtruth are larger than a given threshold (*i.e.*, 0.5). We refer readers to [58] for more details.

3.3 Submitted Trackers

In VisDrone-SOT2020, we summarize 13 tracking algorithms from 18 institutes, as described in the Appendix A. Many of them are improved based on recent state-of-the-art trackers from major computer vision conferences such as CVPR, ICCV and ECCV. Similar to VisDrone-SOT2018 [57] and VisDrone-SOT2019 [16], each participating algorithm can be submitted for at most 10 times. The performance is determined by the best results of ten submissions. As shown in Fig. 2, for training of tracking algorithms, one can use any existing tracking datasets (*e.g.*, LaSOT [17], TrackingNet [45], UAVDT [14], UAV123 [43], COCO [37], ImageNet DET [49], ImageNet VID, YouTube-BB [47], and GOT-10k [26]) in addition to the sequences from training set of VisDrone-SOT2020. Besides, other video object segmentation datasets are also allowed in the training phase, *e.g.*, YouTube-VOS[1] and DAVIS[2].

[1] https://youtube-vos.org/dataset/vos/.
[2] https://davischallenge.org/.

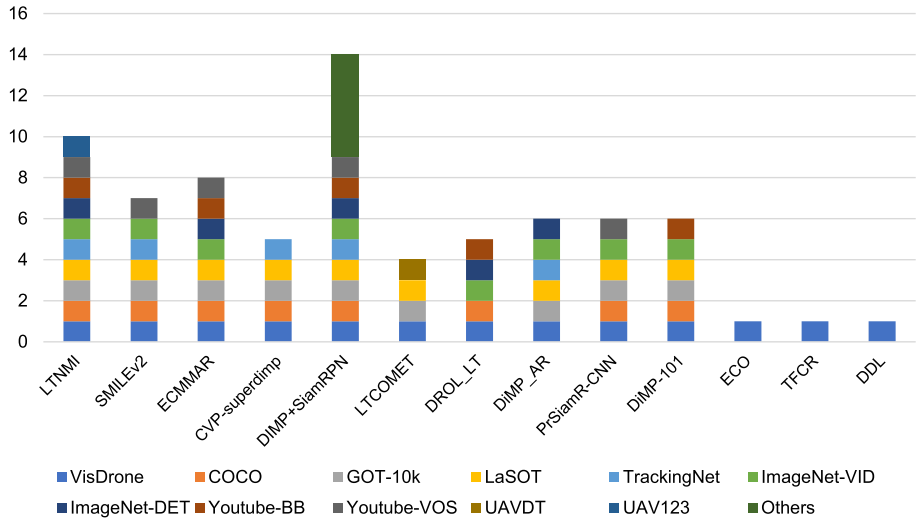


Fig. 2. The training datasets used in the VisDrone-SOT2020 Challenge.

Among the submitted trackers, 7 trackers are inspired by DiMP [5], including SMILEv2 (A.1), LTNMI (A.2), ECMMAR (A.3), CVP-superdimp (A.4), DIMP-SiamRPN (A.7), DiMP_AR (A.8) and DiMP-101 (A.10). One tracker PrSiamR-CNN (A.9) adopts a multi-stage re-detection framework for target tracking. One tracker DROL_LT (A.6) is the variation of Siamese tracking algorithm by introducing an online update strategy to adapt to appearance changes. One tracker LTCOMET (A.5) leverages contextual information for tracking. One tracker ECO (A.11) is based on discriminative correlation filter with deep features. One tracker TFCR (A.12) formulate tracking as regression problem. One tracker DDL (A.13) employs both deep feature and hand-crafted feature for tracking.

4 Results and Analysis

In this section, we evaluate all submitted tracking algorithms on the testing set of VisDrone-SOT2020 dataset and discuss several representative trackers in terms of short-term tracking, long-term tracking and different visual attributes, respectively. Finally, several potential research directions are concluded.

4.1 Overall Performance

We report the tracking results of each submitted tracking algorithm using precision and success scores, as shown in Fig. 4. As shown in Fig. 4, in term of precision score, SMILEv2 achieves the best results of 0.911. The approach of SMILEv2 is an ensemble of different tracking algorithms including three recent state-of-the-art DiMP, SiamMask and SORT. Besides, motion information obtained by

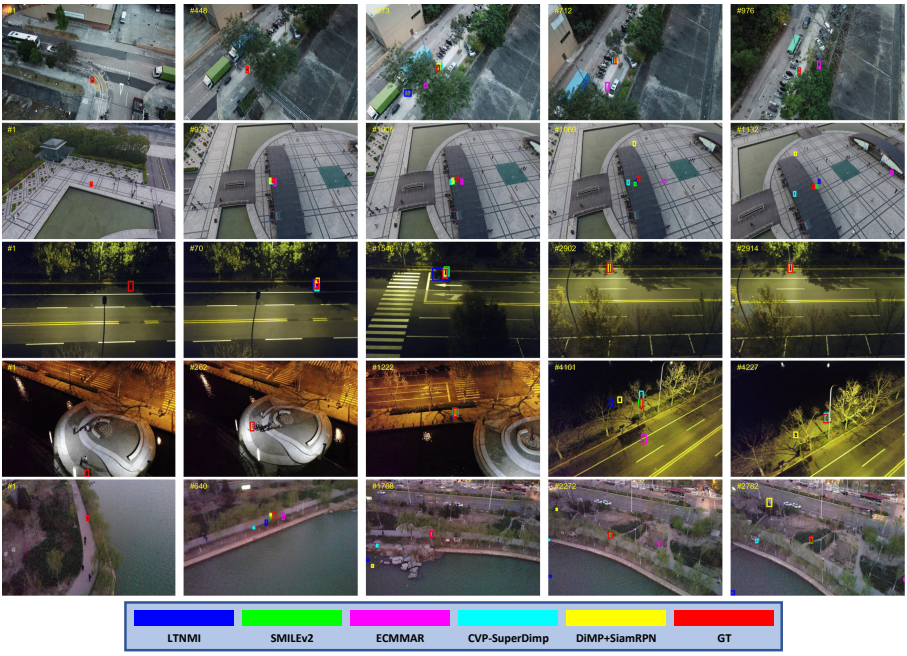


Fig. 3. Tracking results of top five trackers on VisDrone-SOT2020. Best viewed in color and zooming-in. (Color figure online)

optical flow is leveraged to improve the robustness of SMILEv2. LTNMI obtains the second best performance of 0.889 precision score. Similar to SMILEv2, the method of LTNMI utilizes several state-of-the-art tracking approaches DiMP and SiamRPN++ for tracking. In addition, it utilizes an addition detection Siam R-CNN module to re-locate the target object when it is lost. Moreover, an extra image enhancement module is used to improve the image quality for tracking. The method of ECMMAR achieves the third best result of 0.838 precision score. This method utilizes two tracking algorithms to locate the target object. In addition, to get the accurate bounding box of object, it employs a segmentation approach to refine the final tracking result. In term of success score, SMILEv2 and LTNMI both achieve the best results of 0.660. ECMMAR obtains the third best result of 0.630 success score. To better demonstrate these tracking algorithms, we show some qualitative results of the top five trackers on VisDrone-SOT2020 in Fig. 3.

It is worth noting that, compared to VisDrone-SOT2019, the performance of submitted trackers in VisDrone-SOT2020 are significantly improved. In specific, the success scores of the best three trackers in VisDrone-SOT2019 are 0.635, 0.617 and 0.594, respectively. In comparison with these trackers, the top three tracking algorithms in VisDrone-SOT2020 respectively achieve the success scores of 0.660, 0.660 and 0.630, demonstrating better performance.

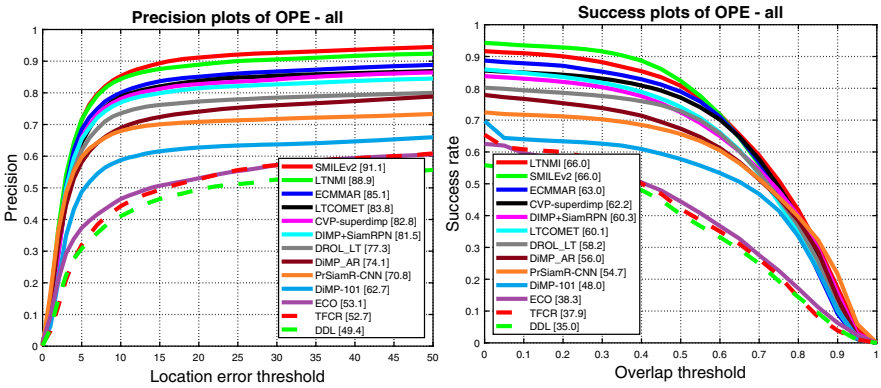


Fig. 4. The success and precision plots of the submitted trackers on all testing videos. The success and precision scores for each tracker are presented in the legend. Best viewed in color and zooming-in. (Color figure online)

The main reason accounting for this is the better baseline trackers used for VisDrone-SOT2020. For example, the most common baseline of the top trackers in VisDrone-SOT2019 is ATOM [8], while it is replaced with a stronger tracker DiMP [5] in VisDrone-SOT2020. In addition, deeper feature representation also helps improve the performance.

4.2 Short-Term Tracking

Following VisDrone-SOT2019, we further perform analysis of short-term tracking performance of different trackers. The results are demonstrated in Fig. 5. The top three trackers are LTNMI, SMILEv2 and LTCOMET. In specific, LTNMI achieves the 0.923 precision score and 0.765 success score. SMILEv2 obtains the 0.906 precision score and 0.734 success score. LTCOMET achieves the 0.889 precision score and 0.711 success score. An interesting observation is that, although SMILEv2 achieves the best overall performance, LTNMI performs more robust on short-term videos. We argue that the use of image enhancement method may help to improve tracking performance in short videos in LTNMI. In addition, LTCOMET outperforms ECMMAR on short-term videos, which shows the importance of contextual information in improving tracking performance.

4.3 Long-Term Tracking

Likewise, we conduct analysis of long-term tracking performance of each tracking algorithm. The results are shown in Fig. 6. The top three trackers are SMILEv2, ECMMAR and LTNMI, respectively. Specifically, SMILEv2 achieves 0.919 precision score and 0.555 success score. ECMMAR obtains 0.841 precision score and 0.544 success score. LTNMI achieves 0.841 precision score and 0.513 success score. We note that SMILEv2 significantly outperforms ECMMAR and LTNMI,

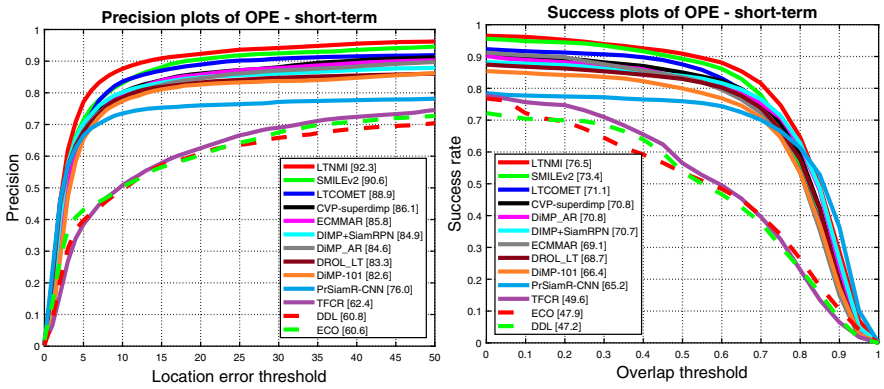


Fig. 5. The success and precision plots of the submitted trackers in term of short-term tracking. The success and precision scores for each tracker are presented in the legend. Best viewed in color and zooming-in. (Color figure online)

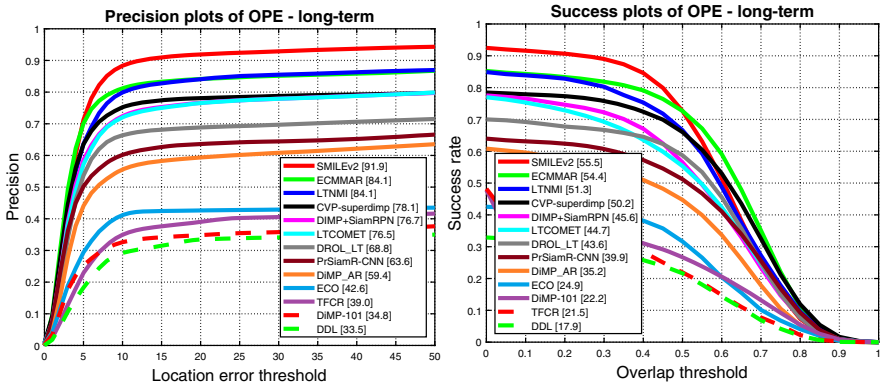


Fig. 6. The success and precision plots of the submitted trackers in term of long-term tracking. The success and precision scores for each tracker are presented in the legend. Best viewed in color and zooming-in. (Color figure online)

which demonstrates the effectiveness of incorporating motion information for long-term tracking.

4.4 Attribute-Based Analysis

In order to further analyze the performance of different tracking algorithms, we show the results of each tracker under all 12 attributes including ARC, BC, CM, FM, FOC, POC, IV, LR, OV, SO, SV and VC. The results are shown in Fig. 7 and 8. In term of precision score, SMILEv2 achieves the best performance on 8 out 12 attributes, including CM, FOC, IV, LR, OV, POC, SO and VC. LTNMI obtains the best result on 4 out of 12 attributes including ARC, BC,

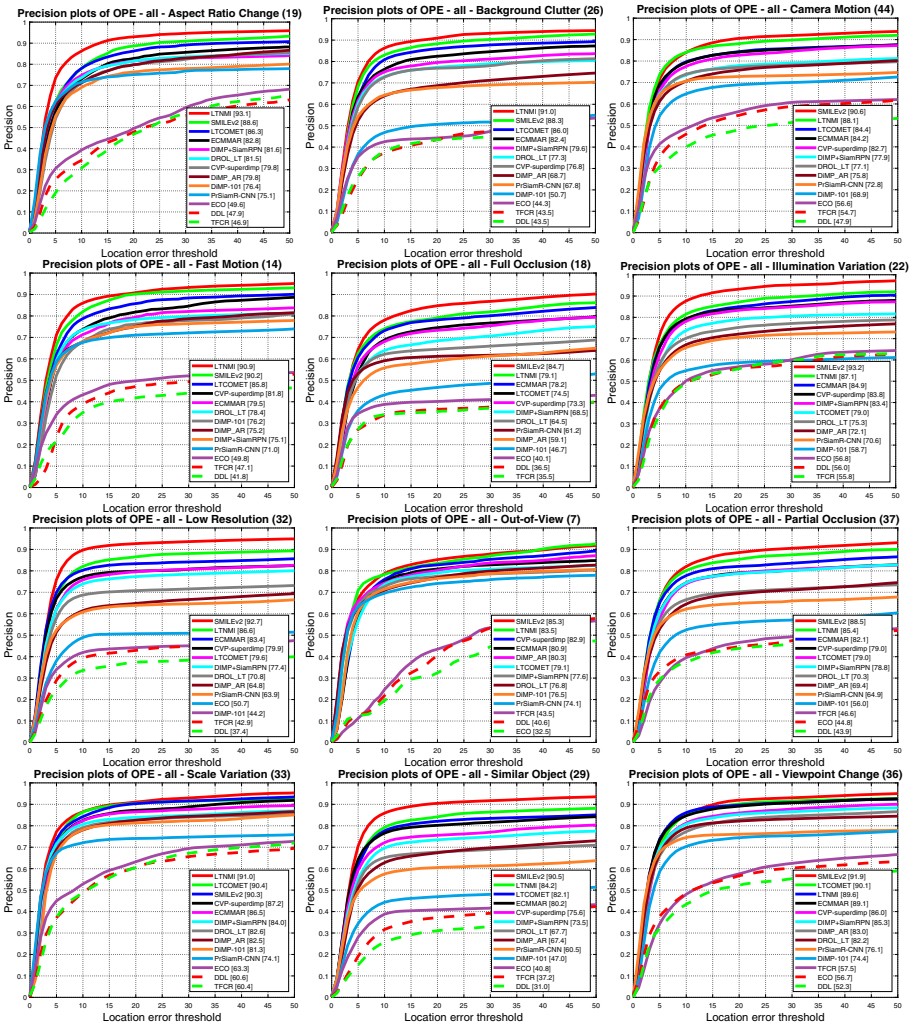


Fig. 7. The precision plots for the submitted trackers in different attributes. The number presented in the title indicates the number of sequences with that attribute. Best viewed in color and zooming-in. (Color figure online)

FM and SV, and it performs competitively on all other attributes. These finds show the effectiveness of applying multiple complementary trackers in tracking performance improvement. In term of success score, on the other hand, LTNMI achieves the best performance on 8 out of 12 attributes including ARC, BC, CM, FM, LTNMI, POC, SV and VC. The method os SMILEv2 obtains the best results on 4 out of 12 attributes including FOC, IV, LR and SO.

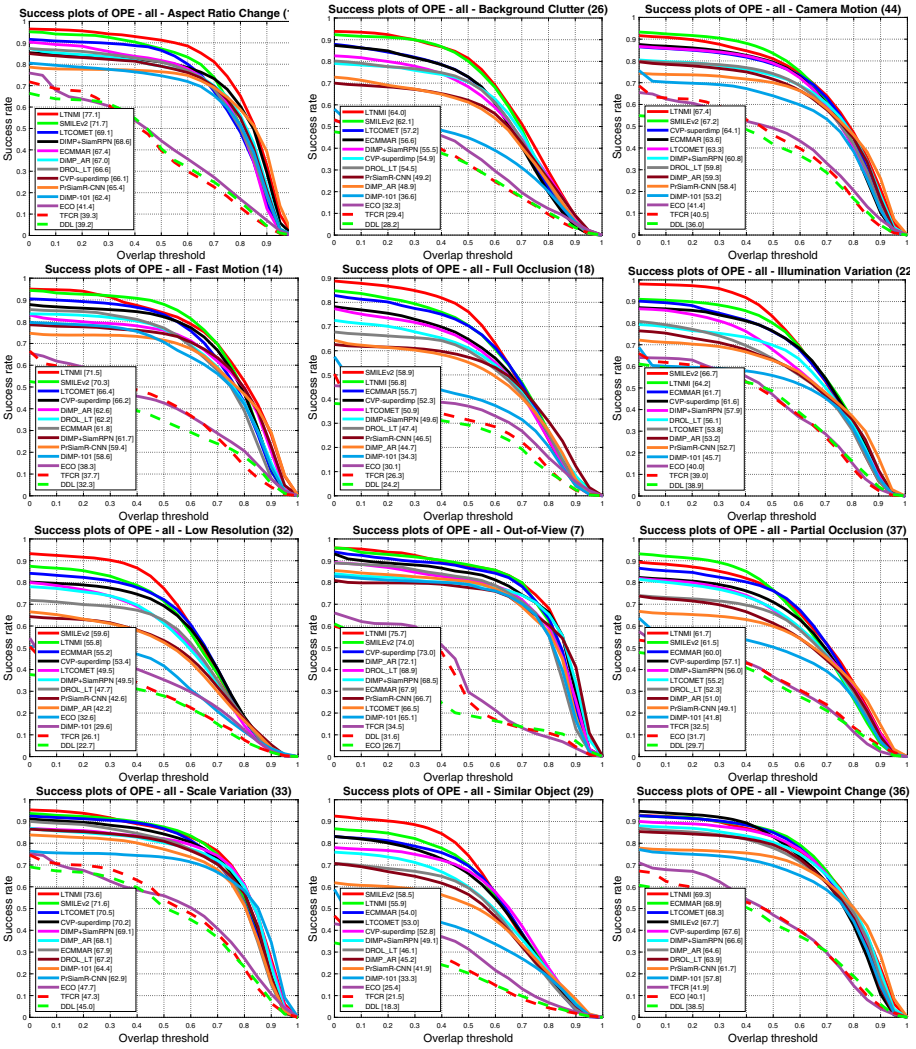


Fig. 8. The success plots for the submitted trackers in different attributes. The number presented in the title indicates the number of sequences with that attribute. Best viewed in color and zooming-in. (Color figure online)

4.5 Discussion

In order to provide guidance for designing robust tracker, we summarize several directions to work on in future research based on the analysis of submitted trackers.

Ensemble of Different Trackers. According to the analysis of submitted trackers, we observe that most of them utilize multiple tracking algorithms to

locate the target, aiming to make full use of the advantage of each tracker for robustness. Especially, the top three trackers SMILEv2, LTNMI and ECMMAR utilize two or three state-of-the-art trackers proposed recently. We believe that this strategy will be widely adopted in the future. However, an interesting direction is how to combine different trackers in an adaptive way.

Integration of Tracking and Detection. In long-term tracking, the target object may frequently disappear and then re-enter the view. In such case, it is necessary to re-detect the target object when it appears again in the view. To this end, a possible solution is to integrate an additional detection model for tracking, as in LTNMI and DROL_LT. Since currently these method often utilize the detector explicitly, it is worth exploring how to build an end-to-end network that integrates both tracking and detection.

Incorporation of Motion Information. Motion information has been crucial for video understanding. In visual tracking, motion information can be used to filter out background distractors, especially for long-term tracking. The method of SMILEv2 effectively leverages such information and achieves the best overall performance.

5 Conclusion

In this paper, we introduce the VisDrone-SOT2020 challenge for drone-based tracking research, in conjunction with ECCV 2020. In specific, the VisDrone-SOT2020 dataset consists of 167 videos with more than 189K frames, aiming to provide a dedicated platform for both training and evaluation of drone tracking algorithm. In addition, we extensively evaluate and discuss 13 submitted tracking algorithms in depth, which serves as baselines on VisDrone-SOT2020 dataset. Based on the analysis of submitted trackers, we share some directions that are worth being explored in the future. By holding the VisDrone-SOT2020 challenge, we hope that it can facilitate the research on drone-based tracking algorithms. For the future, we plan to further expand the dataset and conduct more evaluation of different approaches.

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A Descriptions of Submitted Trackers

In the appendix, we summarize 13 trackers submitted in the VisDrone-SOT2020 Challenge, which are ordered according to the submissions of their final results.

A.1 Strategy and Motion Integrated Long-Term Experts-Version 2 (SMILEv2)

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SMILEv2 is combined with three kind of basic trackers and integrated in our IPIU-tracking framework. In this new framework, we are able to select different trackers in different situations by a semi-automatic way. As shown in Fig. 9, the framework has three parts, which are prediction module, tracking module and fix module. For prediction module, we introduce the Kalman filter and the optical flow method of VCN [61] as the object motion information and camera motion information respectively. For tracking module, we use three trackers including Dimp [5], SiamMask [56], SORT-MOT [4]. For fix module, we first obtain the output of prediction and tracking modules, and then judge the final result.

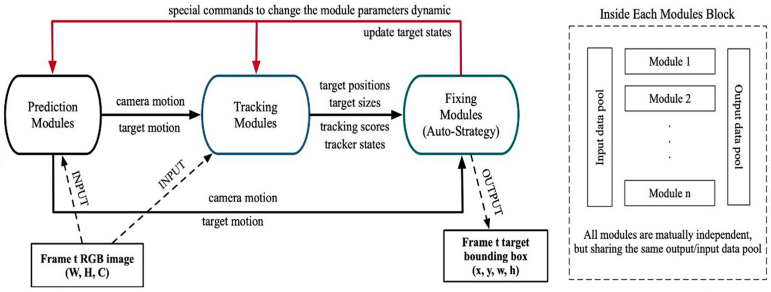


Fig. 9. The framework of SMILEv2.

A.2 Long-Term Tracking with Night-Enhancement and Motion Integrated (LTNMI)

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LTNMI is a combination of ATOM [8], SiamRPN++ [31], Siam-RCNN [54] and Dimp [5]. We combined the ATOM and SiamRPN++ to get a better result, and then our method can give the reliability low limits of the above two systems on the condition of different confidence levels, which makes the systems more reliable respectively as different features play different role in the process of tracking based on their reliability. In addition, we improve the prediction of blurred scenes by using SIFT algorithm to match features. By estimating motion, the regression boxes can continue tracking the target in case of occlusion. When encountering dark or low-resolution scenes, we use threshold judgment and image brightness enhancement processing. We use MBLLEN [40] algorithm to process weak light

enhancement. And then, we use Dimp to get the result of the sequences with weak light enhancement. At last, we use Siam-RCNN to find some lost frames. As a result, when the overlap are of fused result and the result generated by Siam-RCNN is nearly 95%, we conclude that the result generated by Siam-RCNN is better because of accurate detecting bounding box.

A.3 Ensemble of Classification and Matching Models with Alpha-Refine for UAV Tracking (ECMMAR)

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ECMMAR tracker is improved from Dimp [5] and SiamRPN++ [31] with online update module [65]. Dimp performs well in distinguishing distractors, while SiamRPN++ with the re-detection module performs well in detecting target when target disappeared by full occlusion or fast perspective conversion. The main modification are: 1) Develop an interactive mechanism to handle with long-term tracking and improve the robustness. 2) Muti-scale search regions are set to help to re-detect target when full occlusion or fast perspective conversion happened. 3) Use a refinement module [60] to refine the localized bounding box. 4) Employ a low-light image enhancement [62] method to deal with low-light scenes. 5) Fine-tune the superdimp pre-trained model and alpha-refine pre-trained model with visdrone2020 dataset. 6) Motion compensation is used when the camera viewing angle changes greatly. 7) Inertial motion is added when both tracker results are unreliable.

A.4 UAV Tracking with Extra Proposals Based on Corrected Velocity Prediction (CVP-superdimp)

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CVP-superdimp is a robust tracking strategy under the circumstance of UAV tracking, especially for the nerve-wracking problem of fierce camera moving and long-term full occlusion. The base tracker follows [5,9], which contains two modules: object classification module based on DIMP and bounding box regression module based on prDIMP. Our proposed tracking strategy adds a new module of velocity prediction for both short-term and long-term, which can provide additional high-quality proposals for tracker searching in the next frame.

A.5 LTCOMET: Context-Aware IoU-Guided Network for Small Object Tracking (LTCOMET)

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To bridge the gap between aerial views tracking methods and modern trackers, the modified context-aware IoU-guided tracker (LTCOMET) is proposed that exploits the offline reference proposal generation strategy (same as COMET tracker [42]), multitask two-stream network [42], kindling the darkness (KinD) [64], and photo-realistic cascading residual network (PCARN) [1]. The network architecture is the same as [42] without using channel reduction after the multi-scale aggregation and fusion modules (MSAFs). The KinD employs a network for light adjustment and degradation removal, which is employed as a preprocessing of LTCOMET on target patches. Also, the LTCOMET employs the generator network of PCARN to recover high-resolution patches of target and its context from low-resolution ones. Furthermore, the proposed method uses a widowing search strategy when it loses the target. The proposed LTCOMET has been trained on a broad range of tracking datasets while it exploits various photometric and geometric distortions (*i.e.*, data augmentations) to improve the variability of target regions.

A.6 Discriminative and Robust Online Learning for Long Term Siamese Visual Tracking (DROL_LT)

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DROL_LT is based on DROL [65]. DROL proposes an online module with an attention mechanism for offline Siamese networks to extract target-specific features under L2 error. DROL also proposes a filter update strategy adaptive to treacherous background noises for discriminative learning, and a template update strategy to handle large target deformations for robust learning. DROL_LT adds two modules to improve DROL results in long term tracking tasks. (1) A detector is added to help DROL recover the targets, which disappear and appear many times. ROI Align is used to extract the features from mixed offline feature maps with the bounding boxes information from detector. (2) A mechanism is designed to help tracker to decide when to update online classifiers and when to use detectors, which depends on a set of thresholds given from experience.

A.7 Discriminative and Robust Online Learning for Long Term Siamese Visual Tracking (DIMP-SiamRPN)

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DIMP-SiamRPN is improved based on PrDIMP [9] and SiamRPN++ [31]. First, we use the frame numbers to divide videos in the challenge set into long-term videos and short-term videos. The short videos are tested using PrDIMP's hyper-parameter adjustment model to obtain the results. Daytime scenes in the long videos are tested by the SiamRPN++ model. In the SiamRPN++ model, we enlarge the instance size 15 pixels every frame, and the upper limit of the search threshold is 1000. In addition, when the target seems to be lost, we reset the center of search scope to the center of the image. Furthermore, we define a make-up strategy to deal with occlusion. As scenes of night in the long videos, we divide them into strong light scenes and dark scenes according to the light intensity, in which different inference parameters are used.

A.8 Discriminative Model Prediction and Accurate Re-detection for Drone Tracking (DiMP_AR)

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DiMP_AR is based on the DiMP [5] by adding a re-detection module. We use the DiMP tracker as a local tracker to predict target state normally and the RT-MDNet [28] is used as a verifier to verify the prediction of DiMP. If the verification is above a predefined threshold, the normal local tracking is conducted in next frame. Otherwise, the re-detection module will be activated. Firstly, the faster R-CNN detector [48] is used to detect some highly possible target candidates in the whole image of next frame. Then the SiamRPN++ [31] tracker is employed to detect the search regions regarding the possible target candidates. When the target is regained, we switch to local tracking with the tracker DiMP.

A.9 Precise Visual Tracking by Re-detection (PrSiamR-CNN)

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PrSiamR-CNN is modified from recently proposed state-of-the-art single object tracker Siam R-CNN [54] by using extra training data from VisDrone-SOT2020.

A.10 Discriminative Model Prediction with Deeper ResNet-101 (DiMP-101)

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DiMP-101 is based on the DiMP [5] model, adopting deeper ResNet-101 as the backbone. With higher learning capacity of the feature extraction network, the performance of the tracking algorithm has been significantly improved to a new level.

A.11 ECO: Efficient Convolution Operators for Tracking (ECO)

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ECO [7] is a discriminative correlation filter based tracker using deep features. This method introduces a factorized convolution operator and a compact generative model of the training sample distribution to reduce model parameters. In addition, it proposes a conservative model update strategy with improved robustness and reduced complexity. More details can be referred to [7].

A.12 Target-Focusing Convolutional Regression Tracking (TFCR)

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TFCR [63] is a target-focusing convolutional regression (CR) model for visual object tracking tasks. This model uses a target-focusing loss function to alleviate the influence of background noise on the response map of the current tracking image frame, which effectively improves the tracking accuracy. In particular, it can effectively balance the disequilibrium of positive and negative samples by reducing some effects of the negative samples that act on the object appearance model.

A.13 DDL-Tracker (DDL)

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DDL-tracker employs deep layers to extract features. Meanwhile, one HOG detector is trained online. If the tracking result is below a threshold, we use the results by the detector.

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